KITABU QUEST

S2

CHEMISTRY

Chameleon

Cover scene

learners carrying out a safe chemistry observation with a chameleon mascot

- Clear lessons
- Worked examples
- Fun activities
- Real-life problems

GRADE 2

RWA REB-CBC aligned draft

Think - Explore - Achieve

Title Page

KITABU QUEST S2 Chemistry

Rwanda REB-CBC learner book

Mascot: chameleon

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S2 learners study Chemistry one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use safe observations, simple models, data tables, and evidence-based explanations.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

As you finish each topic, return to the success criteria and correct one answer before moving on.

Chemical bonding: Lesson Opener

Learning focus: Chemical bonding

Country link: a safe Rwandan observation, model, data table, or investigation for Chemical bonding.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- relate the nature of bonding to properties of substances

Inquiry questions:
- How can chemical bonding help you solve a real problem in Chemistry?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Chemical bonding: Learn

Chemical bonding explains why atoms join and how bonds affect properties.

Core idea:
- Ionic bonding happens when electrons transfer, often between metals and non-metals.
- Covalent bonding happens when atoms share electrons, usually between non-metals.
- Metallic bonding has metal ions in a sea of delocalized electrons.
- Bonding helps explain melting point, conductivity, solubility, and hardness.
- Dot-and-cross or particle diagrams make bonding easier to compare.

Key words:
- Chemical bonding
- Chemical
- bonding
- concept
- observation
- evidence
- safety
- conclusion

Chemical bonding: Worked Example

Model for Chemical bonding: Compare sodium chloride and a simple covalent molecule.

1. Sodium transfers one electron to chlorine, forming ions.
2. Opposite charges attract, so an ionic bond forms.
3. In a covalent molecule, atoms share electrons instead of transferring them.
4. Ionic compounds often conduct electricity when molten or dissolved; simple covalent substances usually do not.

Conclusion: the type of bond helps explain properties.

Chemical bonding: Vocabulary And Study Skill

Use these words and study moves before you practise Chemical bonding. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Chemical bonding
- Chemical
- bonding
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Chemical bonding without a clear example, method, or evidence.

Chemical bonding: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Chemical bonding.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Chemical bonding: Practice And Correction

Practice for Chemical bonding:

Main outcome: relate the nature of bonding to properties of substances.

1. Complete a table comparing ionic, covalent, and metallic bonding.
2. Draw a simple electron-transfer or sharing model.
3. Match each bond type with one property such as conductivity, melting point, or hardness.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Chemical bonding: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Chemical bonding.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Chemical bonding: Outcome Check

Outcome check for Chemical bonding: Relate the nature of bonding to properties of substances.

1. Compare ionic and covalent bonding in one table.
2. State whether electrons are transferred or shared.
3. Link one bond type to one property.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in properties of elements in the: Lesson Opener

Learning focus: Trends in properties of elements in the

Country link: a safe Rwandan observation, model, data table, or investigation for Trends in properties of elements in the.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- describe the trends and patterns in properties of elements in groups and periods

Inquiry questions:
- How can trends in properties of elements in the help you solve a real problem in Chemistry?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in properties of elements in the: Learn

Trends in properties of elements in the is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:
- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:
- Trends in properties of elements in the
- Trends
- properties
- elements
- concept
- observation
- evidence
- safety

Trends in properties of elements in the: Worked Example

Investigation model for Trends in properties of elements in the: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Trends in properties of elements in the: Vocabulary And Study Skill

Use these words and study moves before you practise Trends in properties of elements in the. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Trends in properties of elements in the
- Trends
- properties
- elements
- concept
- observation
- evidence
- safety

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Trends in properties of elements in the without a clear example, method, or evidence.

Trends in properties of elements in the: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Trends in properties of elements in the.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Trends in properties of elements in the: Practice And Correction

Practice for Trends in properties of elements in the:

Main outcome: describe the trends and patterns in properties of elements in groups and periods.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Trends in properties of elements in the: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Trends in properties of elements in the.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Trends in properties of elements in the: Outcome Check

Outcome check for Trends in properties of elements in the: Describe the trends and patterns in properties of elements in groups and periods.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Water pollution: Lesson Opener

Learning focus: Water pollution

Country link: a safe Rwandan observation, model, data table, or investigation for Water pollution.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- assess the causes and effects of water pollution and suggest ways of control
- develop awareness of the dangers of polluted water

Inquiry questions:
- How can water pollution help you solve a real problem in Chemistry?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Water pollution: Learn

Water pollution is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:
- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:
- Water pollution
- Water
- pollution
- concept
- observation
- evidence
- safety
- conclusion

Water pollution: Worked Example

Investigation model for Water pollution: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Water pollution: Vocabulary And Study Skill

Use these words and study moves before you practise Water pollution. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Water pollution
- Water
- pollution
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Water pollution without a clear example, method, or evidence.

Water pollution: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Water pollution.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Water pollution: Practice And Correction

Practice for Water pollution:

Main outcome: assess the causes and effects of water pollution and suggest ways of control.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Water pollution: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Water pollution.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Water pollution: Outcome Check

Outcome check for Water pollution: Assess the causes and effects of water pollution and suggest ways of control.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Water pollution: Outcome Check

Outcome check for Water pollution: Develop awareness of the dangers of polluted water.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Effective ways of waste management: Lesson Opener

Learning focus: Effective ways of waste management

Country link: a safe Rwandan observation, model, data table, or investigation for Effective ways of waste management.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- transform waste materials into different useful materials e.g. fuel (briquettes) and fertilisers (composted manure)

Inquiry questions:
- How can effective ways of waste management help you solve a real problem in Chemistry?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Effective ways of waste management: Learn

Effective ways of waste management is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:
- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:
- Effective ways of waste management
- Effective
- ways
- waste
- management
- concept
- observation
- evidence

Effective ways of waste management: Worked Example

Investigation model for Effective ways of waste management: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Effective ways of waste management: Vocabulary And Study Skill

Use these words and study moves before you practise Effective ways of waste management. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Effective ways of waste management
- Effective
- ways
- waste
- management
- concept
- observation
- evidence

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Effective ways of waste management without a clear example, method, or evidence.

Effective ways of waste management: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Effective ways of waste management.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Effective ways of waste management: Practice And Correction

Practice for Effective ways of waste management:

Main outcome: transform waste materials into different useful materials e.g. fuel (briquettes) and fertilisers (composted manure).

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Effective ways of waste management: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Effective ways of waste management.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Effective ways of waste management: Outcome Check

Outcome check for Effective ways of waste management: Transform waste materials into different useful materials e.g. fuel (briquettes) and fertilisers (composted manure).

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Categories of chemical reactions: Lesson Opener

Learning focus: Categories of chemical reactions

Country link: a safe Rwandan observation, model, data table, or investigation for Categories of chemical reactions.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- differentiate between the types of chemical reactions

Inquiry questions:
- How can categories of chemical reactions help you solve a real problem in Chemistry?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Categories of chemical reactions: Learn

Categories of chemical reactions is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:
- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:
- Categories of chemical reactions
- Categories
- chemical
- reactions
- concept
- observation
- evidence
- safety

Categories of chemical reactions: Worked Example

Investigation model for Categories of chemical reactions: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Categories of chemical reactions: Vocabulary And Study Skill

Use these words and study moves before you practise Categories of chemical reactions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Categories of chemical reactions
- Categories
- chemical
- reactions
- concept
- observation
- evidence
- safety

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Categories of chemical reactions without a clear example, method, or evidence.

Categories of chemical reactions: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Categories of chemical reactions.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Categories of chemical reactions: Practice And Correction

Practice for Categories of chemical reactions:

Main outcome: differentiate between the types of chemical reactions.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Categories of chemical reactions: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Categories of chemical reactions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Categories of chemical reactions: Outcome Check

Outcome check for Categories of chemical reactions: Differentiate between the types of chemical reactions.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Preparation of salts and identification of ions: Lesson Opener

Learning focus: Preparation of salts and identification of ions

Country link: a safe Rwandan observation, model, data table, or investigation for Preparation of salts and identification of ions.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- prepare a salt from suitable starting materials and identify cations and anions in a solution

Inquiry questions:
- How can preparation of salts and identification of ions help you solve a real problem in Chemistry?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Preparation of salts and identification of ions: Learn

Preparation of salts and identification of ions is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:
- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:
- Preparation of salts and identification of ions
- Preparation
- salts
- identification
- ions
- concept
- observation
- evidence

Preparation of salts and identification of ions: Worked Example

Investigation model for Preparation of salts and identification of ions: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Preparation of salts and identification of ions: Vocabulary And Study Skill

Use these words and study moves before you practise Preparation of salts and identification of ions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Preparation of salts and identification of ions
- Preparation
- salts
- identification
- ions
- concept
- observation
- evidence

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Preparation of salts and identification of ions without a clear example, method, or evidence.

Preparation of salts and identification of ions: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Preparation of salts and identification of ions.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Preparation of salts and identification of ions: Practice And Correction

Practice for Preparation of salts and identification of ions:

Main outcome: prepare a salt from suitable starting materials and identify cations and anions in a solution.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Preparation of salts and identification of ions: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Preparation of salts and identification of ions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Preparation of salts and identification of ions: Outcome Check

Outcome check for Preparation of salts and identification of ions: Prepare a salt from suitable starting materials and identify cations and anions in a solution.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Preparation and classification of oxides: Lesson Opener

Learning focus: Preparation and classification of oxides

Country link: a safe Rwandan observation, model, data table, or investigation for Preparation and classification of oxides.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- prepare oxides and classify them based on their properties

Inquiry questions:
- How can preparation and classification of oxides help you solve a real problem in Chemistry?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Preparation and classification of oxides: Learn

Preparation and classification of oxides is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:
- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:
- Preparation and classification of oxides
- Preparation
- classification
- oxides
- concept
- observation
- evidence
- safety

Preparation and classification of oxides: Worked Example

Investigation model for Preparation and classification of oxides: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Preparation and classification of oxides: Vocabulary And Study Skill

Use these words and study moves before you practise Preparation and classification of oxides. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Preparation and classification of oxides
- Preparation
- classification
- oxides
- concept
- observation
- evidence
- safety

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Preparation and classification of oxides without a clear example, method, or evidence.

Preparation and classification of oxides: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Preparation and classification of oxides.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Preparation and classification of oxides: Practice And Correction

Practice for Preparation and classification of oxides:

Main outcome: prepare oxides and classify them based on their properties.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Preparation and classification of oxides: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Preparation and classification of oxides.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Preparation and classification of oxides: Outcome Check

Outcome check for Preparation and classification of oxides: Prepare oxides and classify them based on their properties.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Electrolytes and non- electrolytes: Lesson Opener

Learning focus: Electrolytes and non- electrolytes

Country link: a safe Rwandan observation, model, data table, or investigation for Electrolytes and non- electrolytes.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- distinguish between non-electrolytes, weak electrolytes and strong electrolytes

Inquiry questions:
- How can electrolytes and non- electrolytes help you solve a real problem in Chemistry?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Electrolytes and non- electrolytes: Learn

Electrolytes and non- electrolytes is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:
- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:
- Electrolytes and non- electrolytes
- Electrolytes
- electrolytes
- concept
- observation
- evidence
- safety
- conclusion

Electrolytes and non- electrolytes: Worked Example

Investigation model for Electrolytes and non- electrolytes: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Electrolytes and non- electrolytes: Vocabulary And Study Skill

Use these words and study moves before you practise Electrolytes and non- electrolytes. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Electrolytes and non- electrolytes
- Electrolytes
- electrolytes
- concept
- observation
- evidence
- safety
- conclusion

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Electrolytes and non- electrolytes without a clear example, method, or evidence.

Electrolytes and non- electrolytes: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Electrolytes and non- electrolytes.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Electrolytes and non- electrolytes: Practice And Correction

Practice for Electrolytes and non- electrolytes:

Main outcome: distinguish between non-electrolytes, weak electrolytes and strong electrolytes.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Electrolytes and non- electrolytes: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Electrolytes and non- electrolytes.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Electrolytes and non- electrolytes: Outcome Check

Outcome check for Electrolytes and non- electrolytes: Distinguish between non-electrolytes, weak electrolytes and strong electrolytes.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Properties of organic compounds and uses of alkanes: Lesson Opener

Learning focus: Properties of organic compounds and uses of alkanes

Country link: a safe Rwandan observation, model, data table, or investigation for Properties of organic compounds and uses of alkanes.

Progression focus: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:
- compare the properties of organic and inorganic compounds and explain the uses of alkanes in daily life

Inquiry questions:
- How can properties of organic compounds and uses of alkanes help you solve a real problem in Chemistry?

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Properties of organic compounds and uses of alkanes: Learn

Properties of organic compounds and uses of alkanes is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:
- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:
- Properties of organic compounds and uses of alkanes
- Properties
- organic
- compounds
- uses
- alkanes
- concept
- observation

Properties of organic compounds and uses of alkanes: Worked Example

Investigation model for Properties of organic compounds and uses of alkanes: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Properties of organic compounds and uses of alkanes: Vocabulary And Study Skill

Use these words and study moves before you practise Properties of organic compounds and uses of alkanes. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:
- Properties of organic compounds and uses of alkanes
- Properties
- organic
- compounds
- uses
- alkanes
- concept
- observation

Study skill:
1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Properties of organic compounds and uses of alkanes without a clear example, method, or evidence.

Properties of organic compounds and uses of alkanes: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Properties of organic compounds and uses of alkanes.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Properties of organic compounds and uses of alkanes: Practice And Correction

Practice for Properties of organic compounds and uses of alkanes:

Main outcome: compare the properties of organic and inorganic compounds and explain the uses of alkanes in daily life.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:
A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Properties of organic compounds and uses of alkanes: Challenge And Remediation

Use this page after practice.

If you need support:
1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Properties of organic compounds and uses of alkanes.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:
1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Properties of organic compounds and uses of alkanes: Outcome Check

Outcome check for Properties of organic compounds and uses of alkanes: Compare the properties of organic and inorganic compounds and explain the uses of alkanes in daily life.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:
- evidence is specific
- safety is considered
- conclusion uses the evidence

Chemical bonding: Review Clinic 1

Use this review clinic to strengthen Chemistry without copying earlier answers.

1. Choose one idea from Chemical bonding that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Trends in properties of elements in the: Review Clinic 2

Use this review clinic to strengthen Chemistry without copying earlier answers.

1. Choose one idea from Trends in properties of elements in the that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Water pollution: Review Clinic 3

Use this review clinic to strengthen Chemistry without copying earlier answers.

1. Choose one idea from Water pollution that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Effective ways of waste management: Review Clinic 4

Use this review clinic to strengthen Chemistry without copying earlier answers.

1. Choose one idea from Effective ways of waste management that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Chemical bonding: Application Workshop 1

Application workshop 1 for Chemical bonding.

Grade move: connect ideas across topics and explain causes, methods, or consequences; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Glossary

Use this glossary to revise important words in Chemistry.

- Chemical bonding: Explain this term using an example from the book, your school, or your community.
- Trends in properties of elements in the: Explain this term using an example from the book, your school, or your community.
- Water pollution: Explain this term using an example from the book, your school, or your community.
- Effective ways of waste management: Explain this term using an example from the book, your school, or your community.
- Categories of chemical reactions: Explain this term using an example from the book, your school, or your community.
- Preparation of salts and identification of ions: Explain this term using an example from the book, your school, or your community.
- Preparation and classification of oxides: Explain this term using an example from the book, your school, or your community.
- Electrolytes and non- electrolytes: Explain this term using an example from the book, your school, or your community.
- Properties of organic compounds and uses of alkanes: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Chemistry answer states the idea, uses correct scientific vocabulary, records observations or data, includes units or labels where needed, and writes a conclusion that follows from evidence. Safety rules always come first. If evidence is missing, repeat the observation safely or explain what evidence would be needed.

Final Project

Create a final project for Chemistry that shows what you can now do independently.

Your project must include:
1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.